

Alumni DB — System Flowcharts

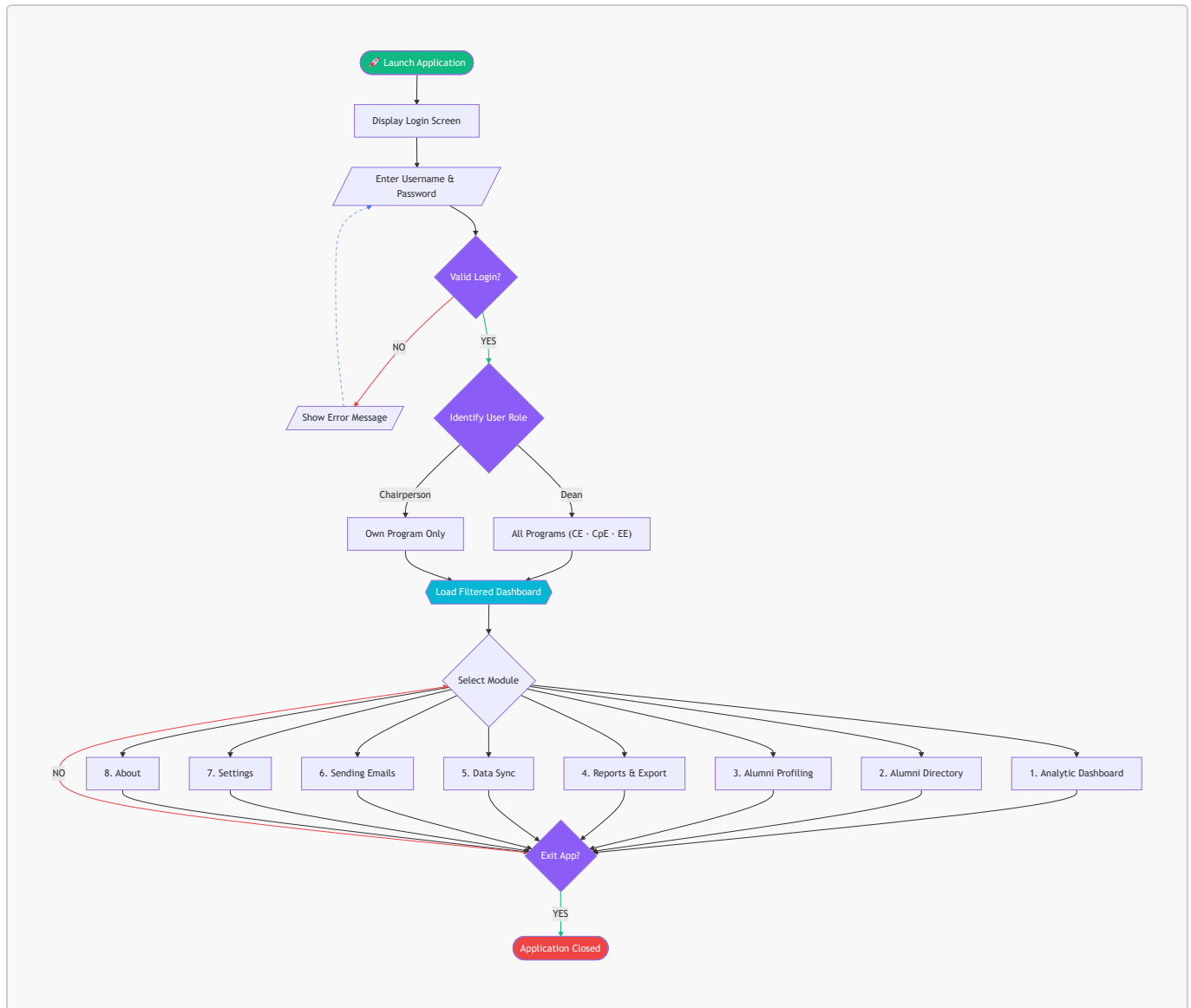
Version: v3.0.0 **Interactive HTML:** [alumni-db-flowcharts-v2-5.html](#) — open in a browser for navigable, color-coded SVG diagrams. **Description:** Modular flowcharts covering all 10 system flows of the Alumni DB Management System, aligned with role-based access for Dean and Program Chairpersons. Updated to reflect dashboard export, ready-made email templates, chairperson sync access, and troubleshooting integration.

Legend

Shape	Meaning
Rounded rectangle / Stadium	Start / End (Terminal)
Rectangle	Process
Diamond / Rhombus	Decision
Parallelogram	Input / Output
Hexagon	Sub-process / Action
Green arrow	YES path
Red arrow	NO / Error path
Blue dashed arrow	Return / loop-back

1. System Overview

Purpose: High-level map of the complete application lifecycle — from launch through login, role-based access, 8 system modules, and exit.



Flow Description

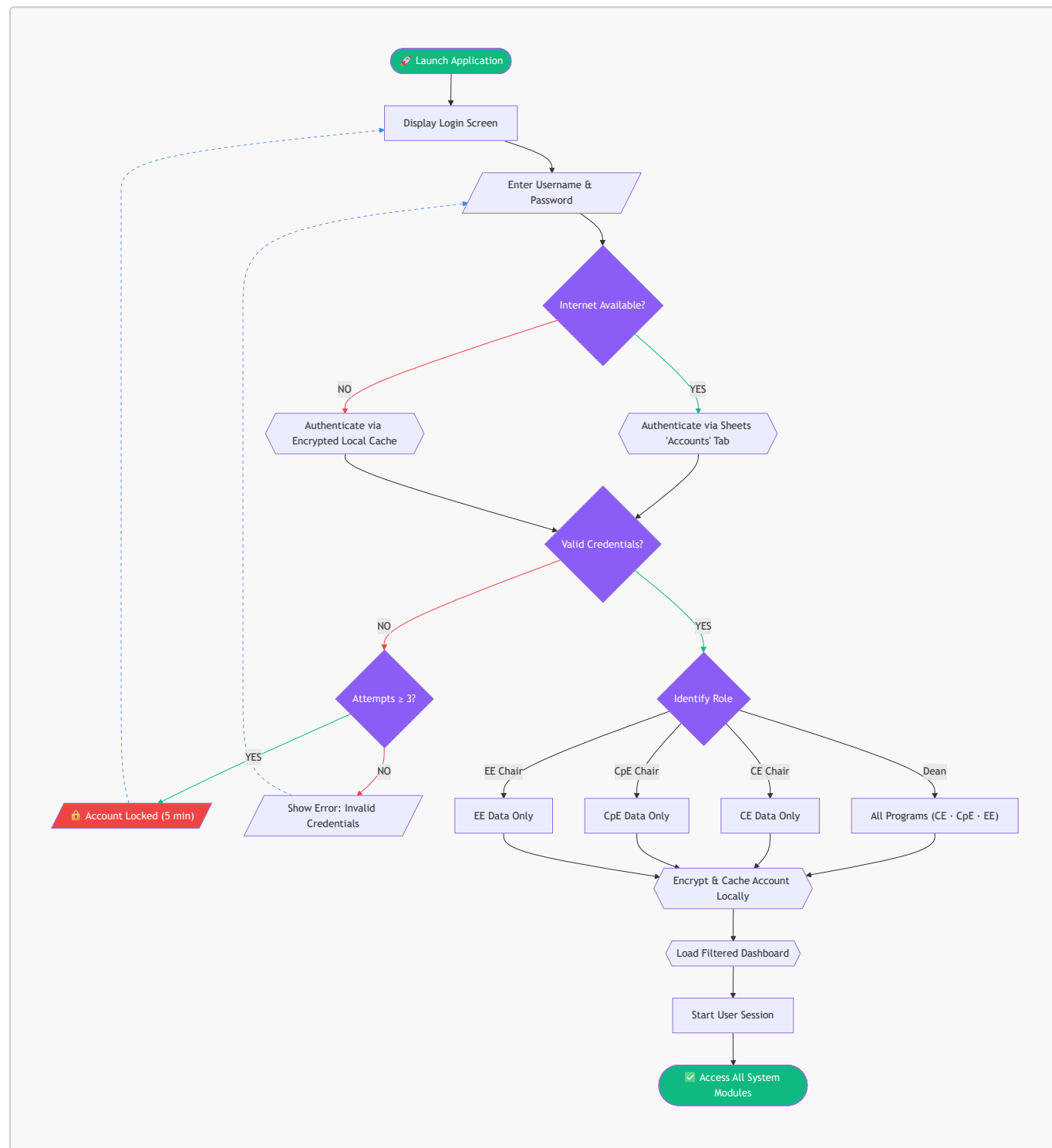
1. User launches the application.
2. The system displays the **Login Screen**.
3. User enters credentials (username & password).
4. **Valid Login?**
 - **NO** → Show error message ("Invalid credentials"). Loop back to enter credentials.
 - **YES** → Proceed to role identification.
5. **Identify User Role?** — Based on the **role** field from the accounts data:
 - **Dean** → Access all programs (CE · CpE · EE), full data scope.
 - **Chairperson** → Access own program data only (CE, CpE, or EE).
6. Both paths merge → Load the **Dashboard (Filtered View)** according to role scope.
7. **Select Module** — User navigates to one of 8 modules.
8. After completing work in any module, the user returns to the Dashboard.
9. **Exit App?**
 - **NO** → Loop back to module selection.
 - **YES** → Application Closed (terminal state).

Additional Details

- **Login Gate:** The v1 "settings gate" is replaced by a login screen. Settings are no longer the entry point — authentication is.
- **Hub-and-Spoke Architecture:** The Dashboard remains the central hub. Every module returns to it.
- **Role-Based Data Scoping:** The role determined at login persists for the entire session and filters all data globally.

2. Login & Role-Based Access

Purpose: Full authentication flow — credential input, validation against accounts stored in the Google Sheets "Accounts" tab (with encrypted offline fallback), attempt lockout, role detection, and session start.



Accounts are stored in a dedicated "**Accounts**" **tab** within the same Google Sheets spreadsheet:

Column	Description
username	Login username (unique)
password	Hashed password
role	dean, ce_chair, cpe_chair, or ee_chair
full_name	Display name
is_active	TRUE / FALSE — only active accounts can log in
created_at	ISO 8601 timestamp
last_login	ISO 8601 timestamp (updated on successful login)

Offline Fallback

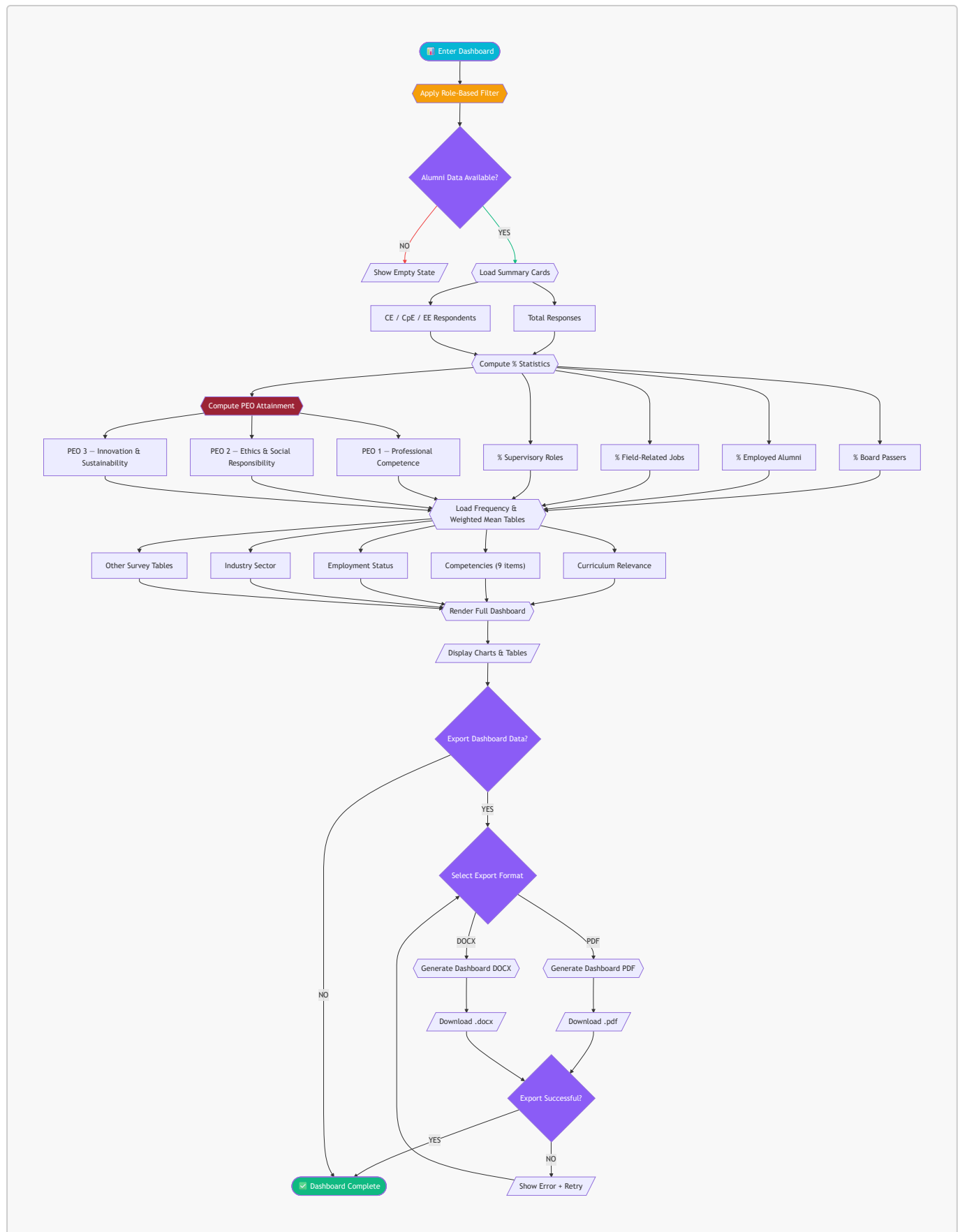
- On the **first successful online login**, the system encrypts and caches the user's account record locally.
- If the internet is unavailable, the system authenticates against this local encrypted cache.
- Cache is refreshed on every successful online login.

Additional Details

- **No Self-Registration:** First account is manually added to the Sheets tab.
- **Lockout Policy:** 3 consecutive failures → 5-minute lockout.
- **Session:** Stored in Zustand `auth.store`, ends on app close.

3. Analytic Dashboard

Purpose: Role-filtered data loading, summary stat cards, percentage metrics, frequency/weighted mean tables, and optional PDF/DOCX export.

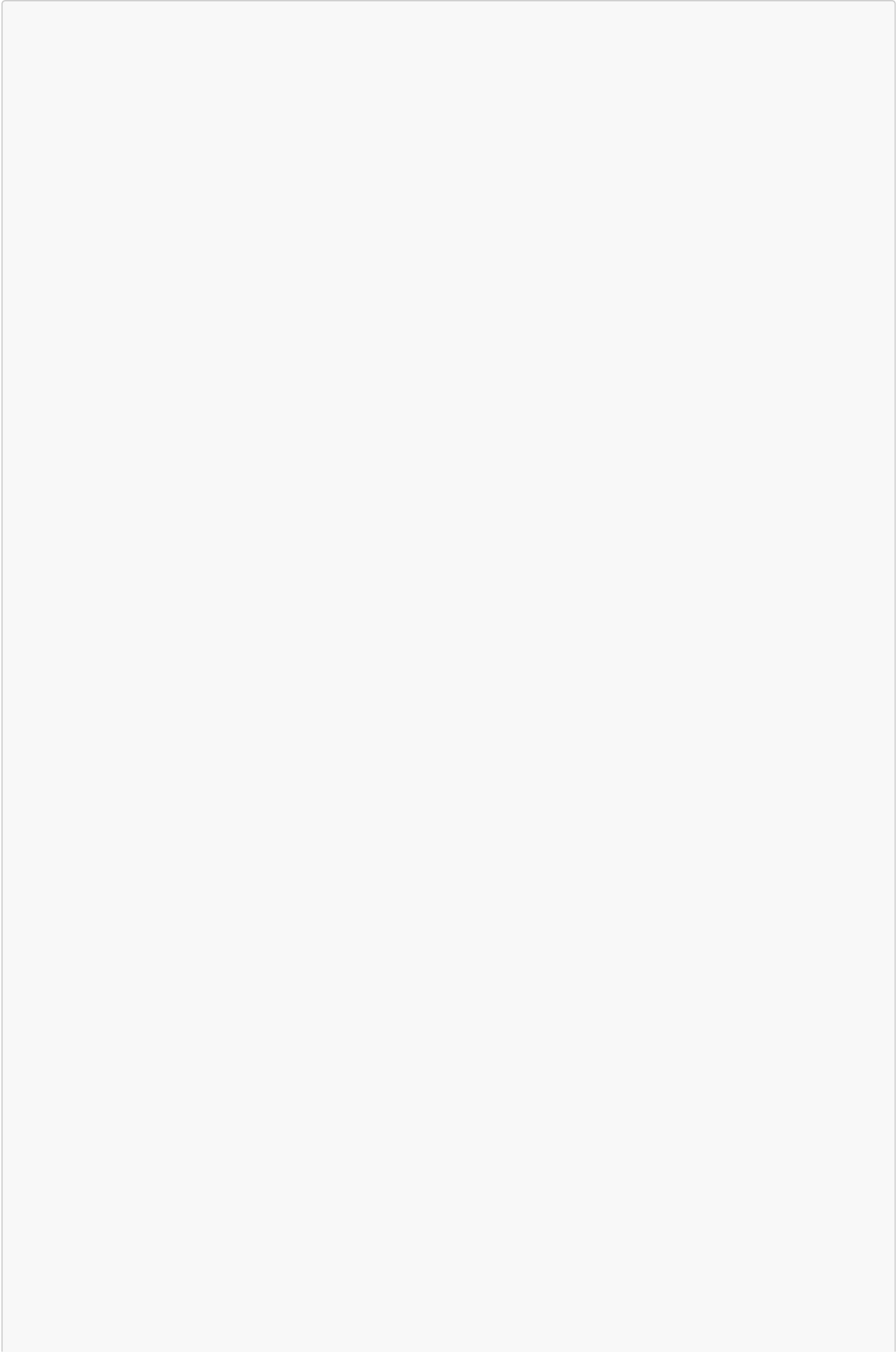


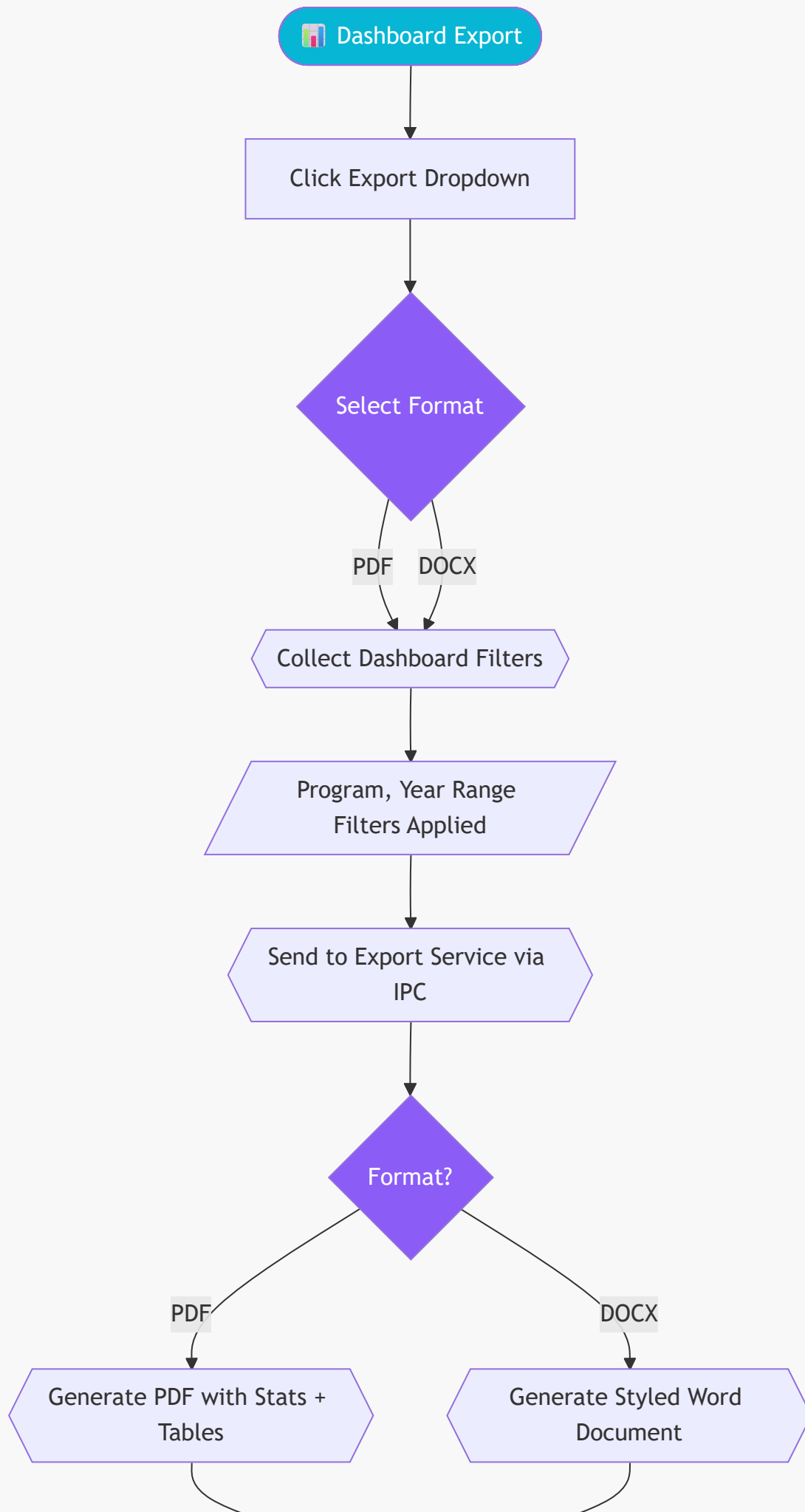
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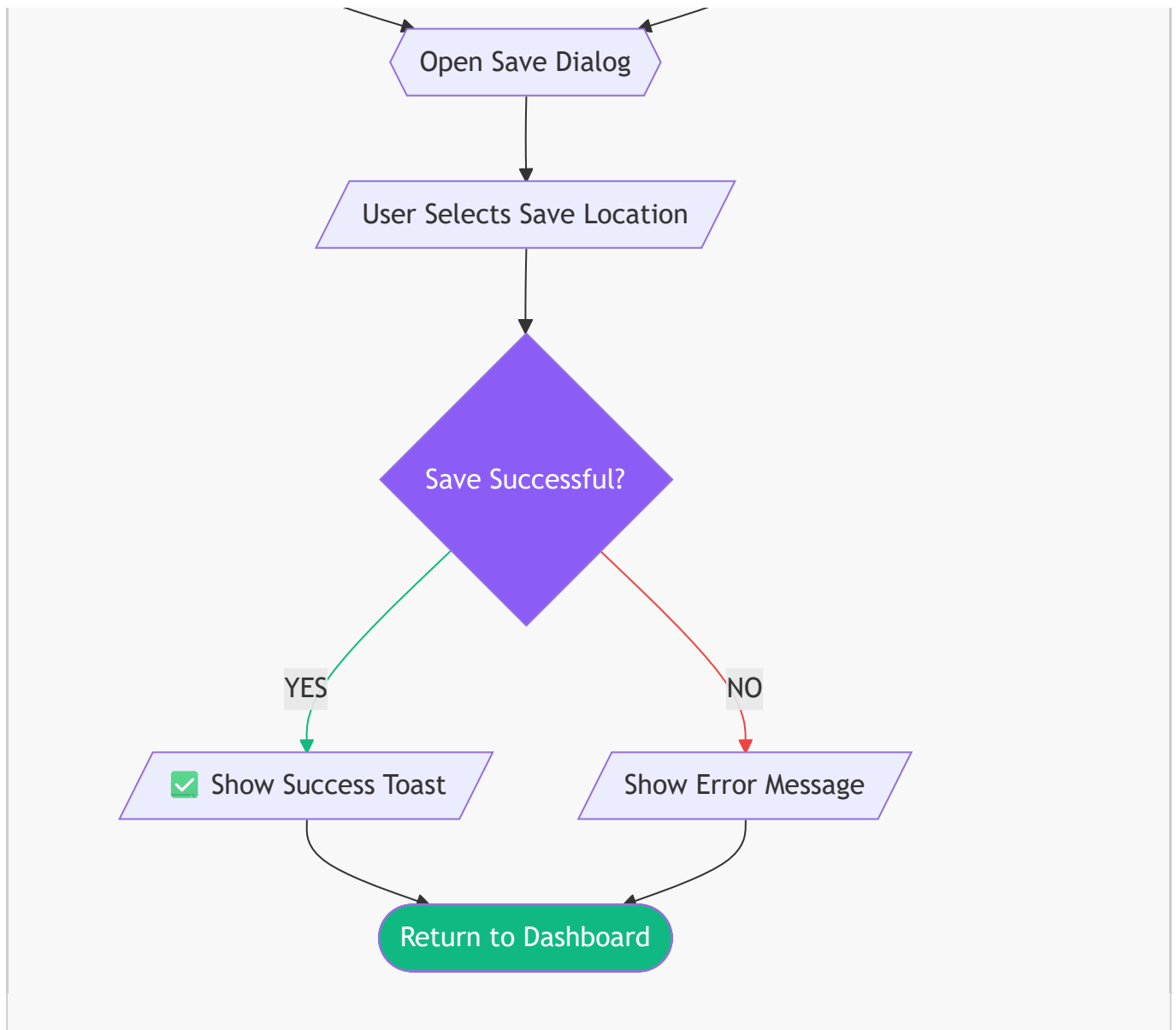
- **Role-Based Filter Applied First:** A Chairperson sees only their own program's statistics.
- **Stat Cards (KPIs):** Total Responses, per-program counts, % Board Passers, % Employed, % Field-Related, % Supervisory.

- **PEO Attainment Cards & Accordion:** Three additional cards display Program Educational Objectives attainment % (PEO 1 Professional Competence, PEO 2 Ethics & Social Responsibility, PEO 3 Innovation & Sustainability) plus a fourth Outcomes (Field-Aligned Employment) card. Clicking any card opens an accordion with per-indicator breakdowns, "How calculated" methodology, "Data fields used", and a graduation-cohort outcome view (recent / mid / established). Computed by **peoService** via the dedicated **peo.repository.ts** SQL constants — separate from the basic KPI percentages.
- **Survey Tables:** Frequency distribution and weighted mean (for Likert-scale data).
- **Dashboard Export:** After viewing charts/tables, user can export the entire dashboard view as PDF or DOCX via the export dropdown button. Filters (program, year range) are passed to the export service.

Sub-Flow 3a: Dashboard Export Flow

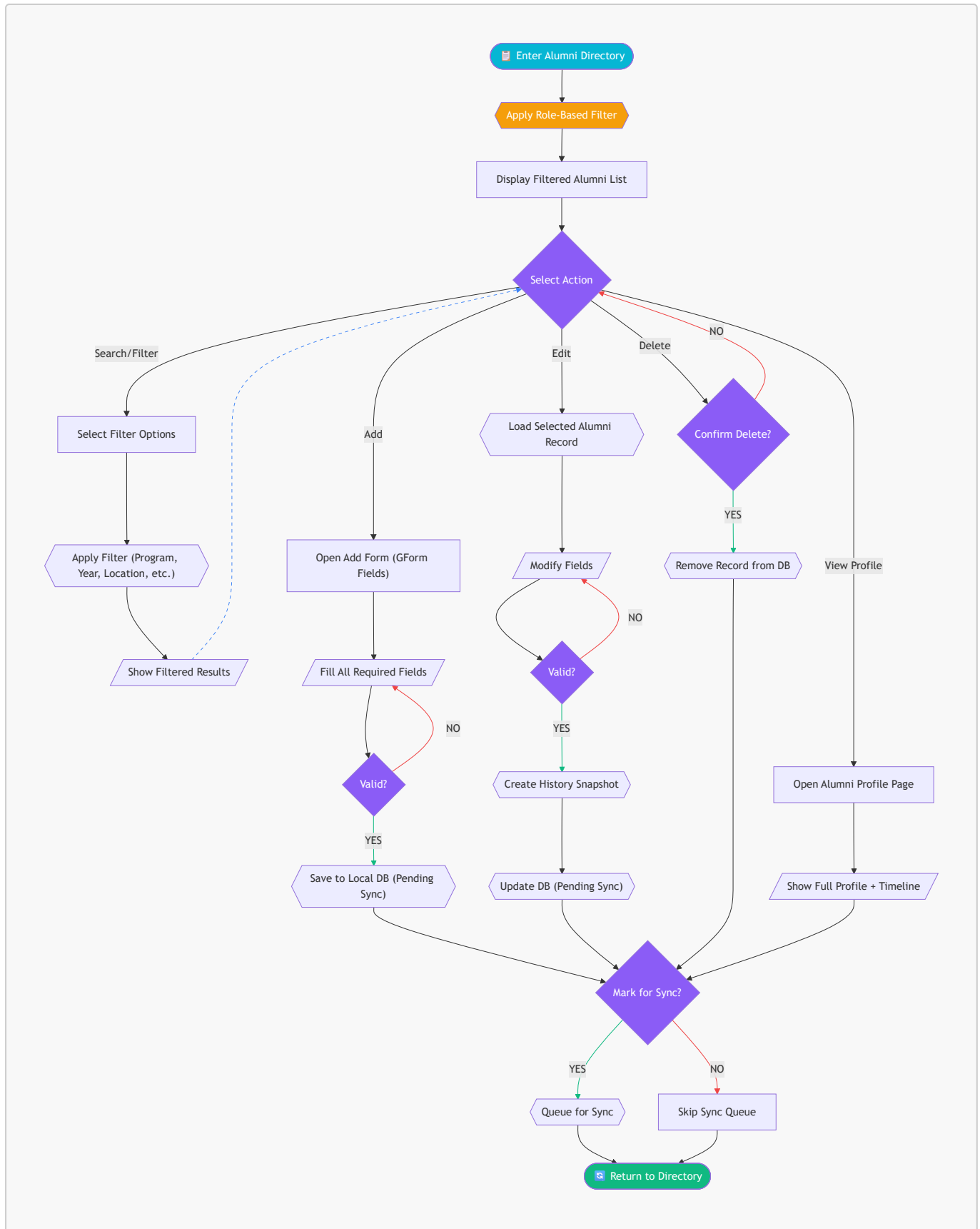






4. Alumni Directory

Purpose: Full CRUD operations with search/filter, GForm-based add, edit, delete, profile view — all role-filtered.

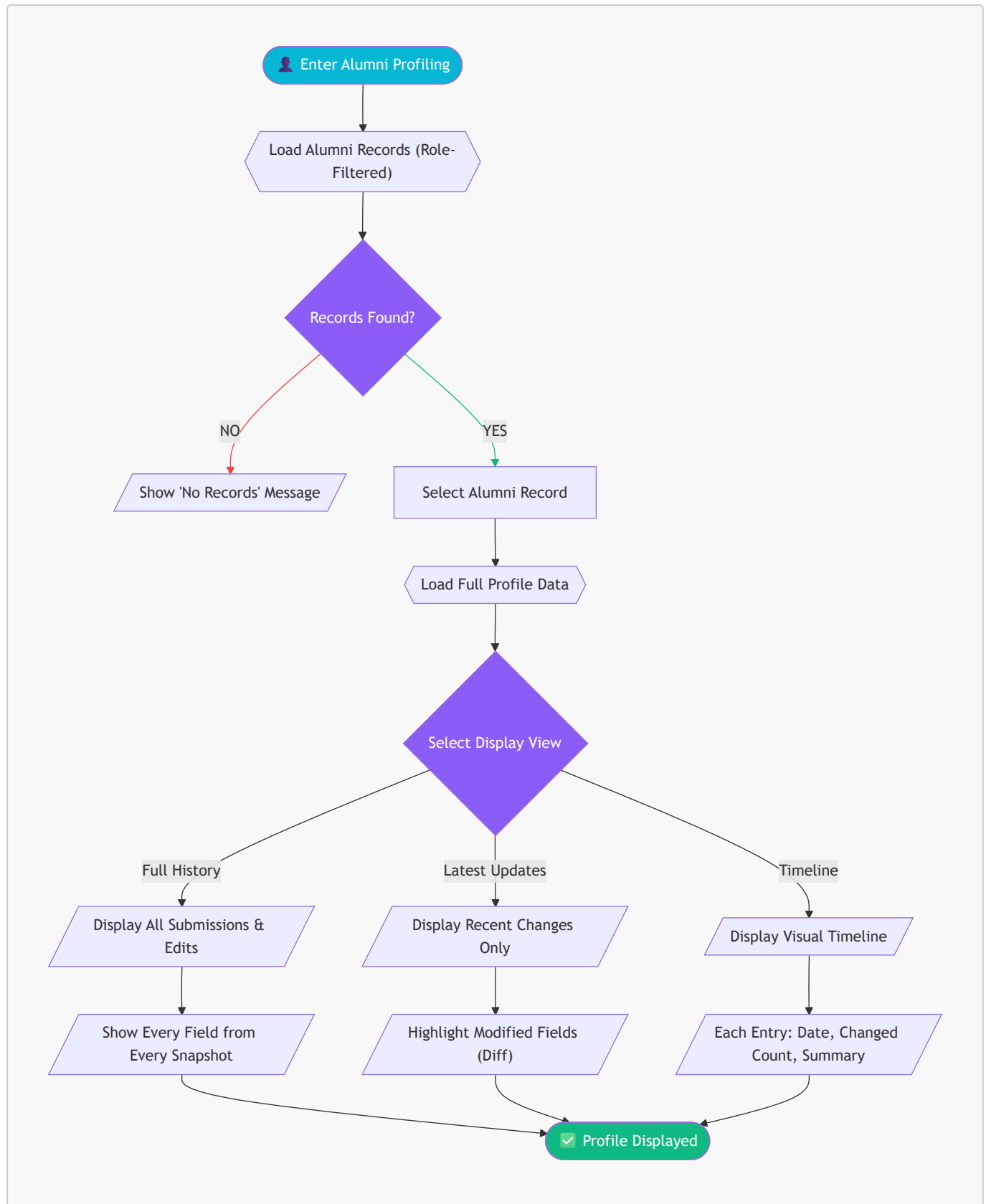


Additional Details

- **GForm Integration:** The "Add" action links to the configured Google Form.
- **Validation Loop:** Both Add and Edit have validation loops.
- **Pending Sync:** All created/edited records are marked `sync_status = 'pending'`.
- **Delete Safety:** Requires explicit confirmation dialog.

5. Alumni Profiling

Purpose: Individual alumni profile view — complete response history, latest update highlights, and chronological timeline.



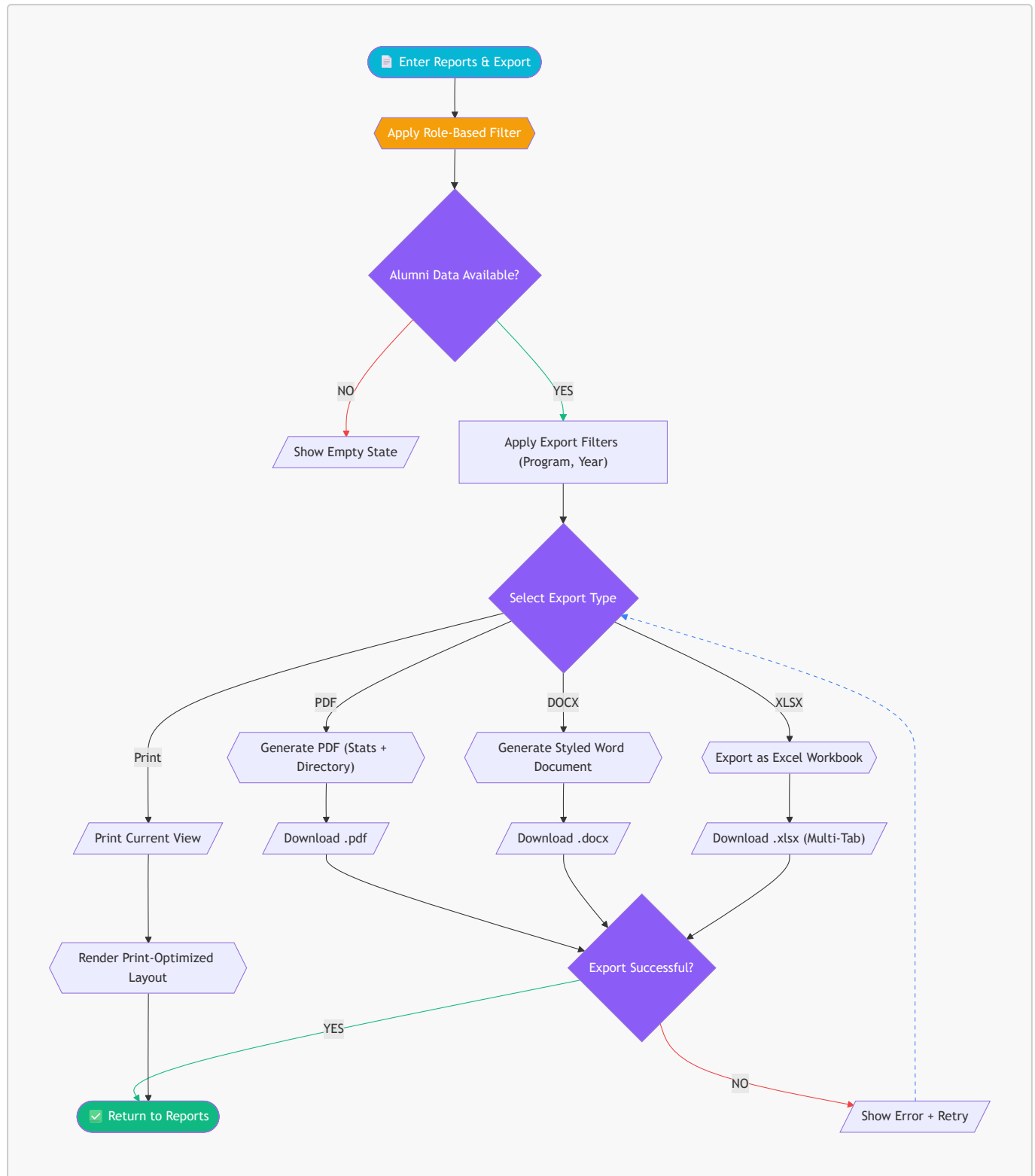
Additional Details

- **History Snapshots:** Every edit creates a JSON snapshot in `alumni_history`.

- **Diff Highlighting:** Latest Updates view shows visual old → new diff indicators.
- **Timeline Navigation:** Click any timeline entry to expand full snapshot.

6. Reports & Export

Purpose: Export alumni data as PDF, DOCX, XLSX — all role-filtered. Includes print option for current view.



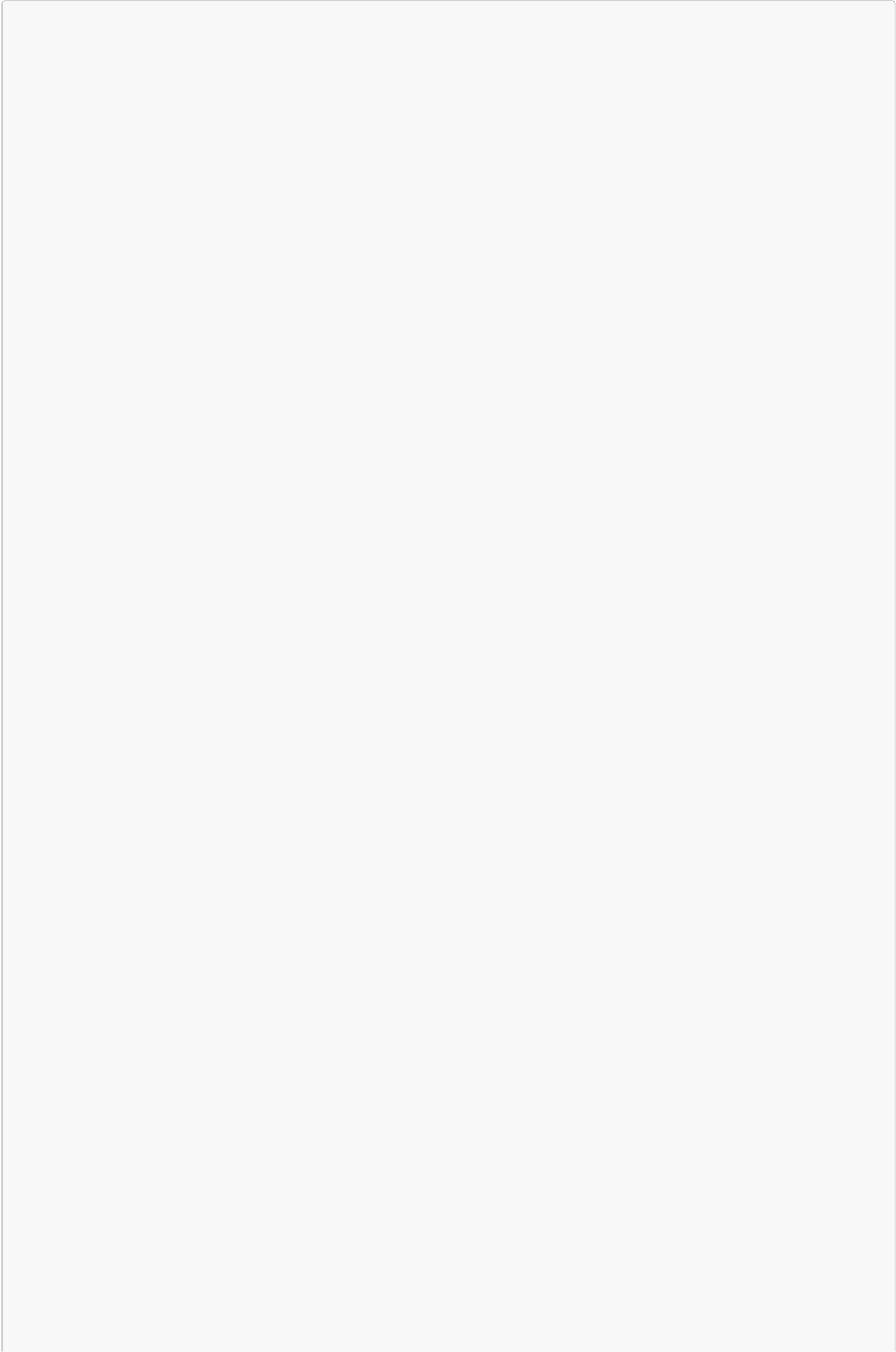
Additional Details

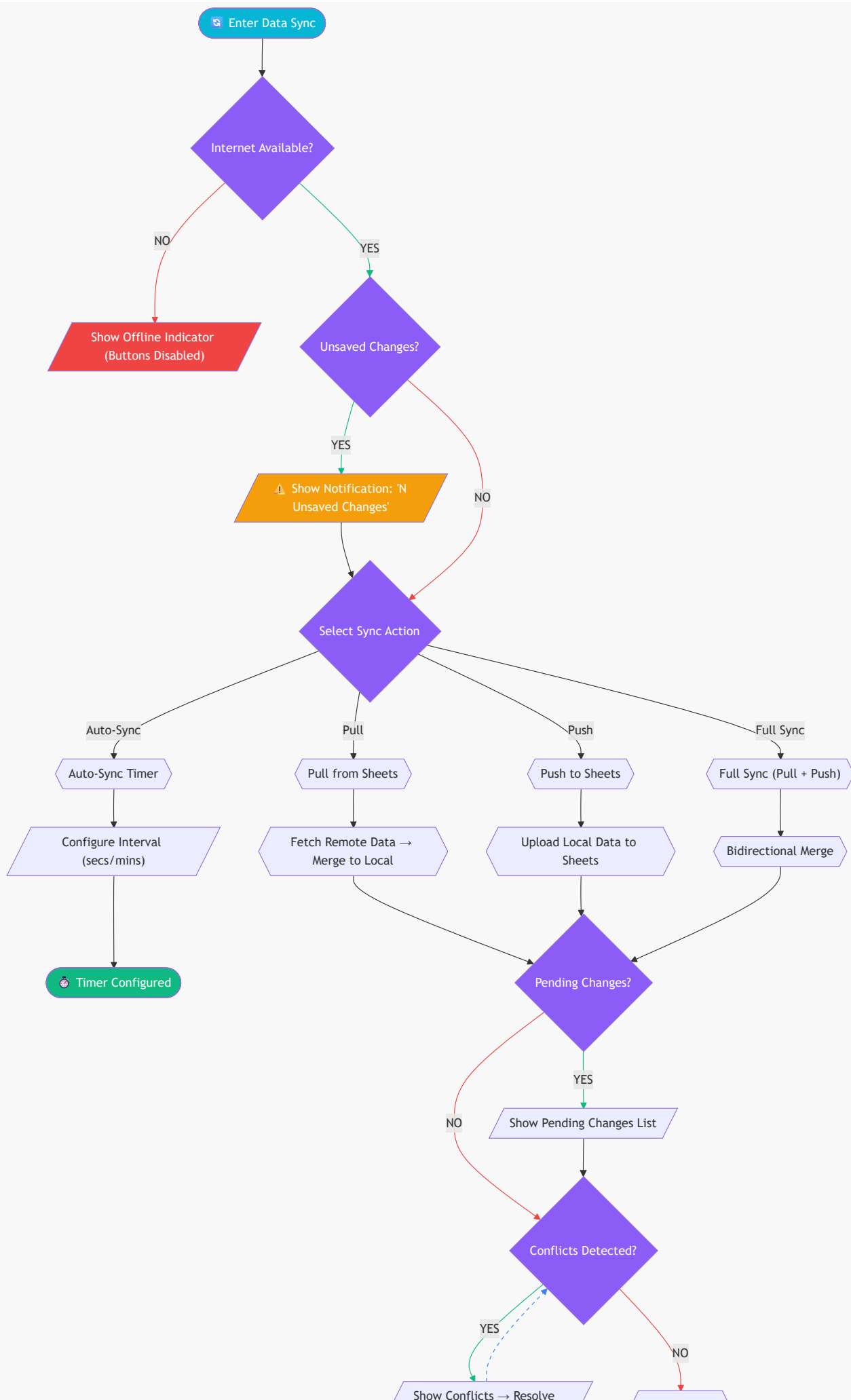
- **Role-Based Filter Applied First:** Chairpersons can only export their own program's data.

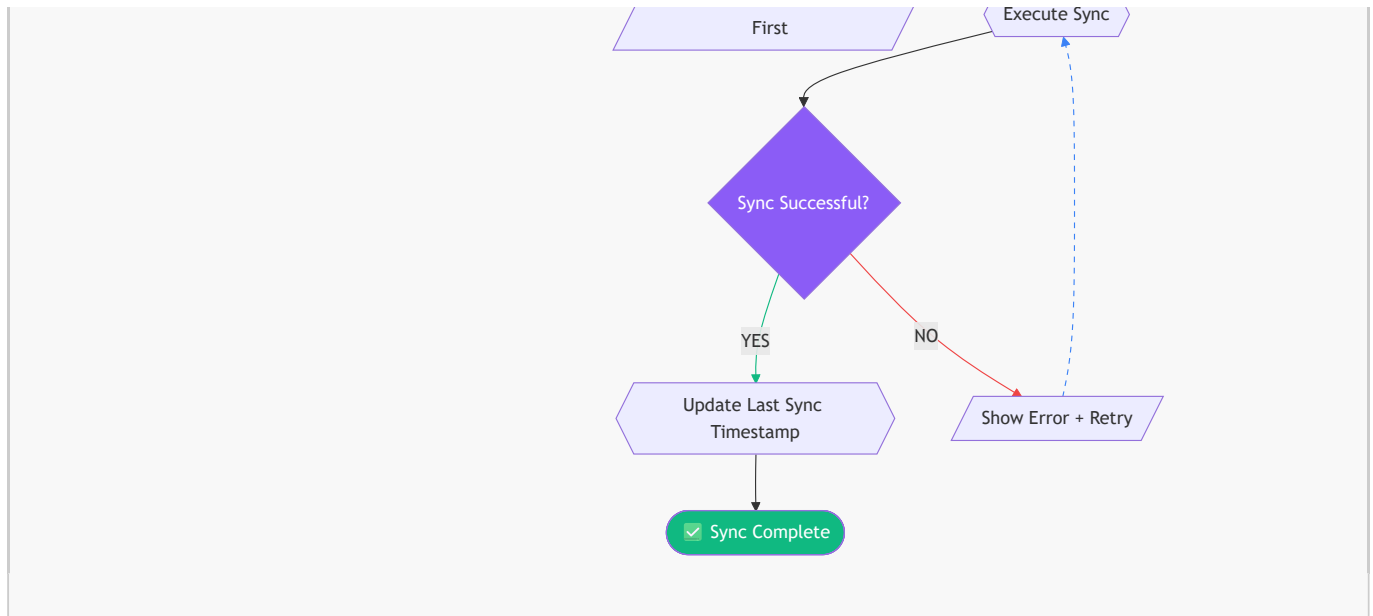
- **3 Export Formats:** PDF, DOCX, XLSX — each as a card in the export grid.
 - **Print Option:** A separate "Print Current View" button triggers `window.print()` with a clean, full-width layout (sidebar/topbar hidden automatically).
 - **Excel Multi-Tab:** Data sheet, Filters Applied tab, Summary Statistics tab.
 - **Preview Branch Removed:** The previous Print Preview/Preview export type has been consolidated into the simpler "Print Current View" action.
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7. Data Synchronization

Purpose: Internet check, unsaved change notification, 4 sync modes, pending changes, conflict resolution. **All roles can access this module — data is scoped to accessible programs.**





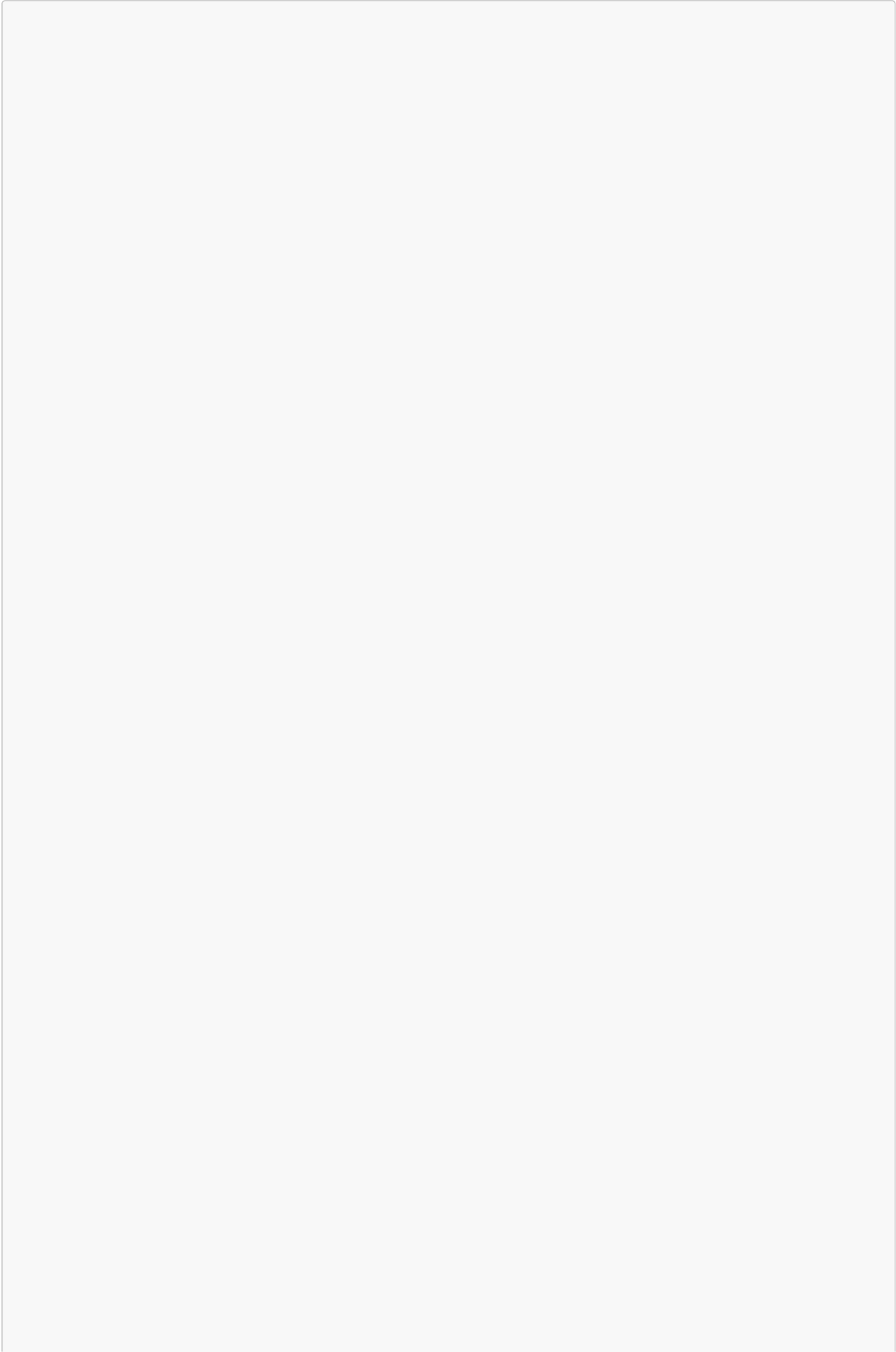


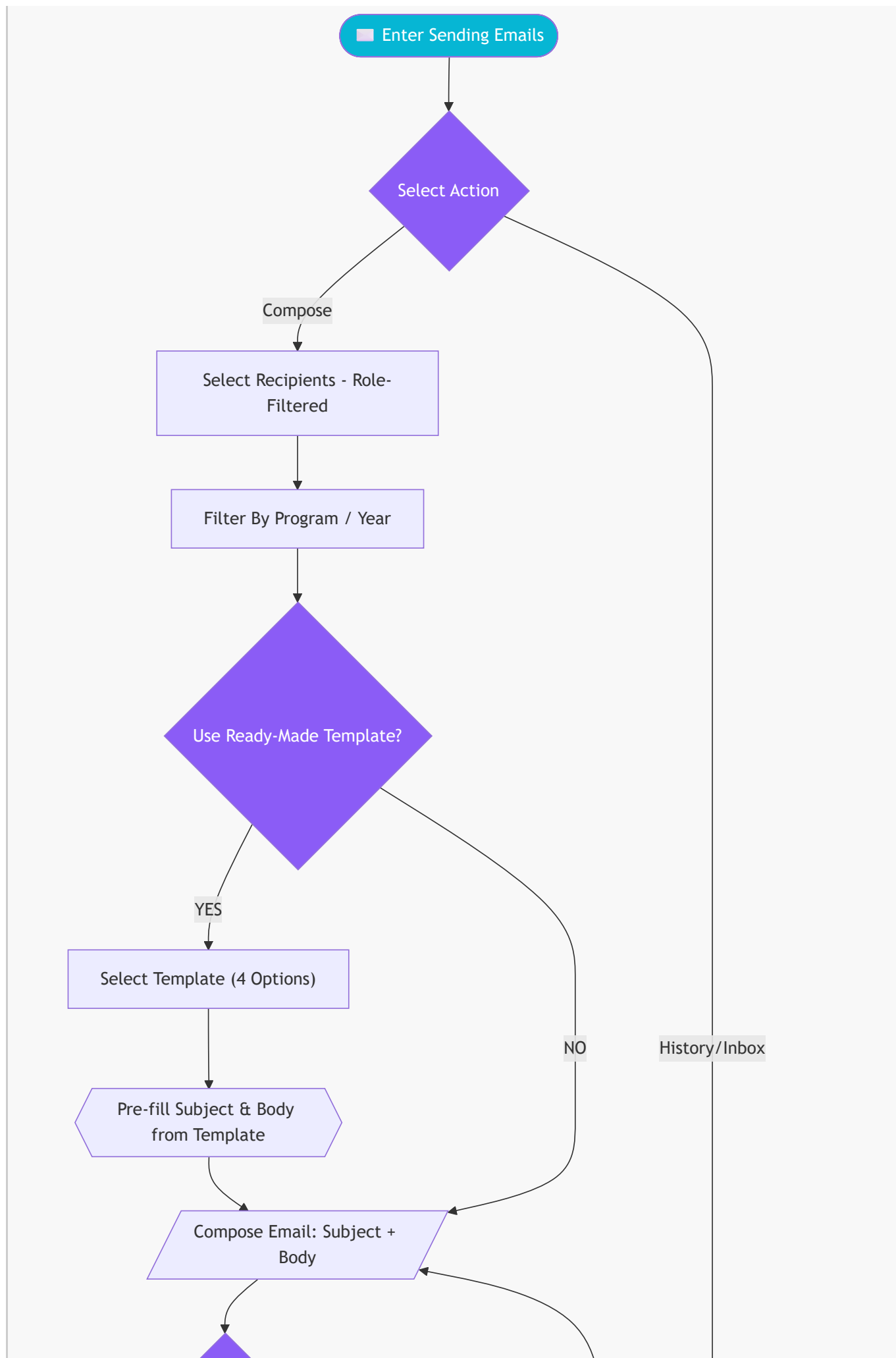
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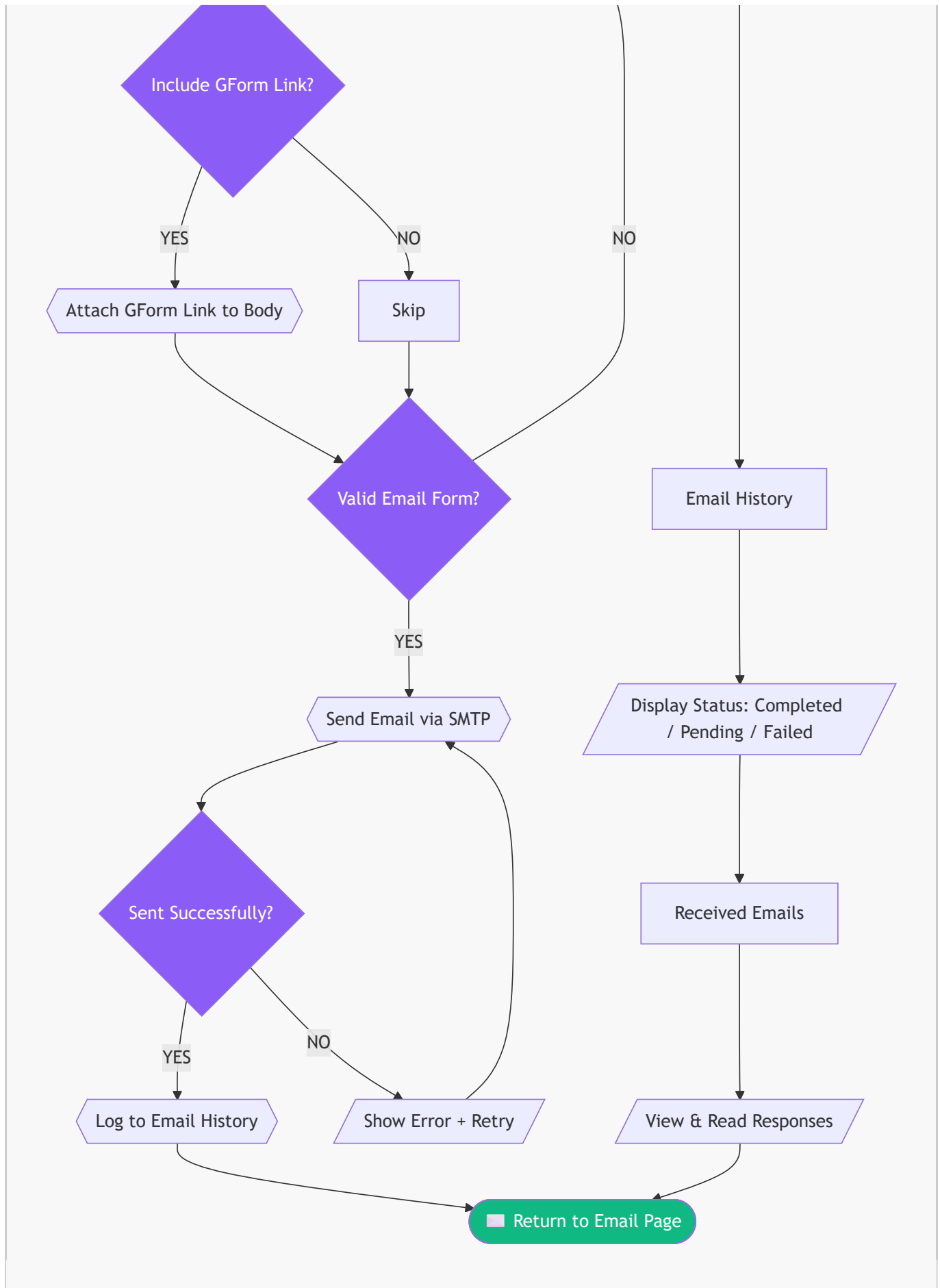
- **Internet Gate:** Sync is impossible without internet.
- **Unsaved Changes Notification:** Alerts user before syncing if local edits exist.
- **Auto-Sync Timer:** Configurable interval. Pauses on conflicts.
- **Conflict Resolution:** Manual — keep local, keep remote, or merge.
- **All Roles Can Sync:** Both Dean and Chairpersons can access the Data Sync module. Data is automatically scoped — Chairpersons only sync their own program's records, while the Dean syncs all programs.

8. Sending Emails

Purpose: Role-filtered recipients, ready-made email templates, GForm link decision, compose, send via SMTP, history tracking.







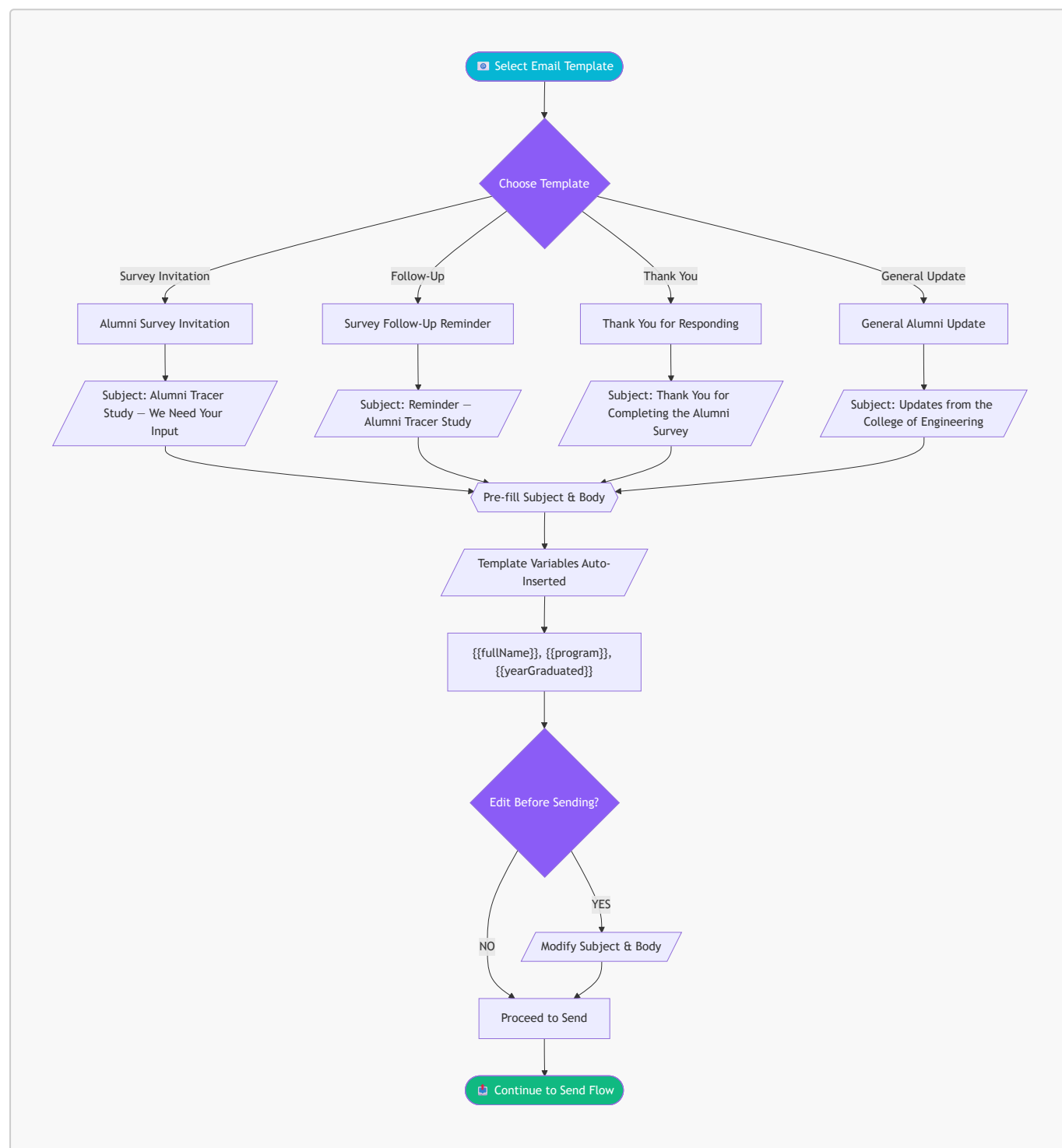
Additional Details

- **Role-Based Recipients:** CE Chairperson can only email CE alumni.

- **Ready-Made Templates:** 4 pre-built email templates (see Sub-Flow 8a below). Templates pre-fill the subject and body with appropriate content and template variables. User can edit before sending.
- **GForm Link Decision:** Optional inclusion for survey reminders.
- **Template Variables:** `{{fullName}}`, `{{program}}`, `{{yearGraduated}}`, etc.
- **Email History:** Every batch send logged with delivery status.

Sub-Flow 8a: Email Template Selection

The compose form includes a **"Use a Ready-Made Template"** button that reveals 4 template cards. Selecting one pre-fills the subject and body fields.

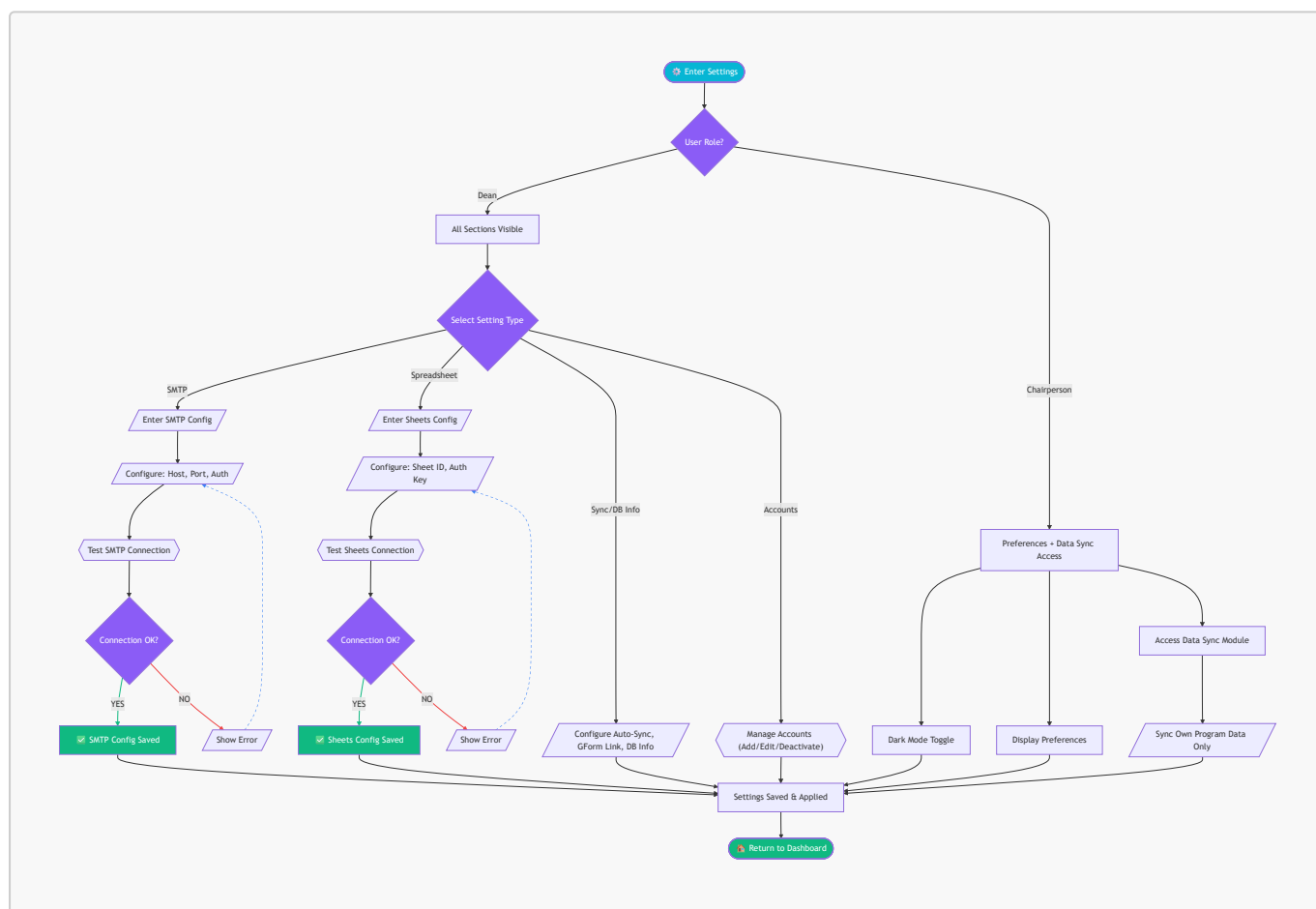


4 Ready-Made Templates:

#	Template Name	Subject Line	Purpose
1	Alumni Survey Invitation	Alumni Tracer Study — We Need Your Input, {{fullName}}!	Initial survey request to alumni
2	Survey Follow-Up Reminder	Reminder: Alumni Tracer Study — Your Response Matters, {{fullName}}	Follow-up for non-responders
3	Thank You for Responding	Thank You for Completing the Alumni Survey, {{fullName}}!	Post-survey appreciation
4	General Alumni Update	Updates from the College of Engineering — {{program}}	Generic communication

9. Settings

Purpose: SMTP + Sheets configuration with test/retry loops, Dean-only management sections, user preferences, and clarification of Chairperson data sync access.



Settings Sections by Role Access

Section	Dean	Chairperson
SMTP Configuration	✓ Manage	✗ Hidden
Spreadsheet Connection	✓ Manage	✗ Hidden

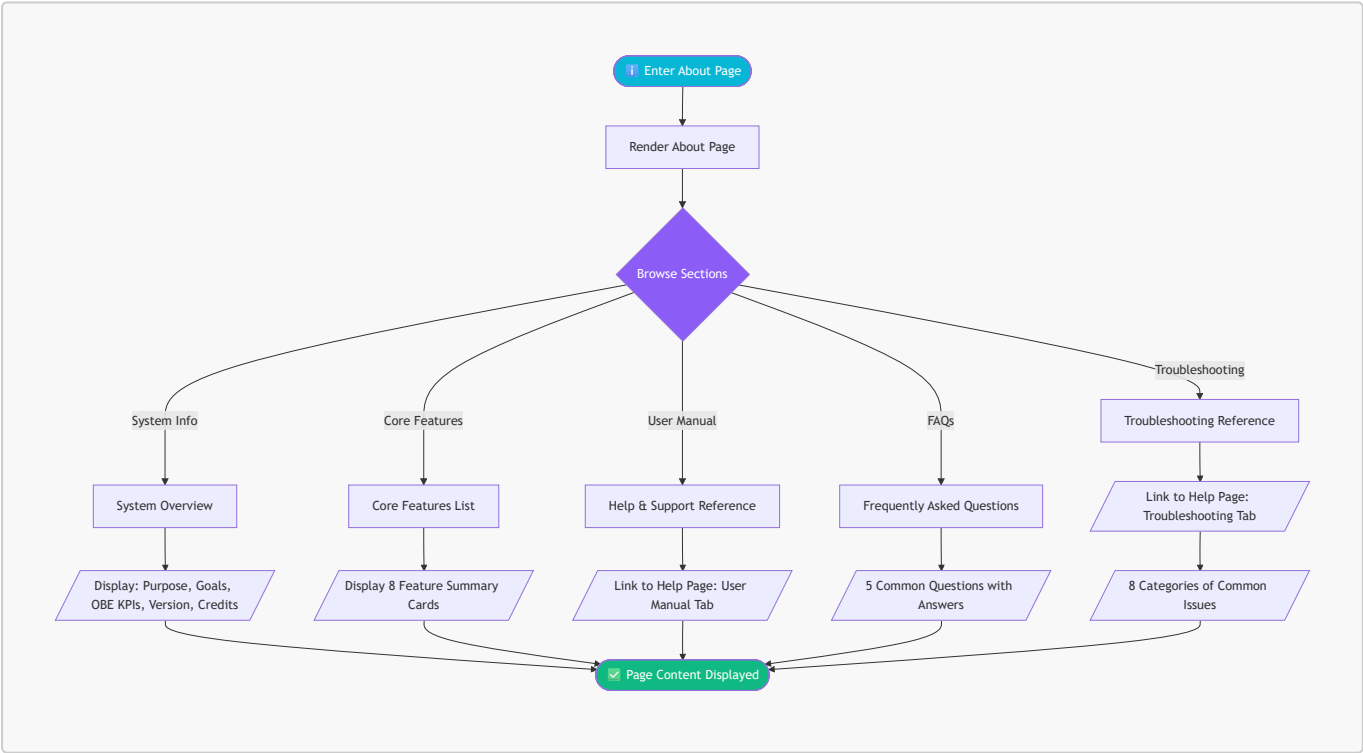
Section	Dean	Chairperson
Sync & Database Info	✔ Manage	✖ Hidden
Accounts Management	✔ Manage	✖ Hidden
User Preferences	✔ Manage	✔ Manage
Data Sync Module (separate route)	✔ All Programs	✔ Own Program Only

Additional Details

- **Chairperson Sync Access:** Chairpersons can perform sync operations (Pull/Push/Full Sync) through the **Data Sync module** (`/sync` route) for their own program's data. The Settings sync configuration (auto-sync interval, GForm link, etc.) remains Dean-only. The `/sync` route has no role guard — data scoping is applied at the query level via `accessiblePrograms`.
- **Section Visibility:** For Chairpersons, the SMTP, Spreadsheet, Sync/DB Info, and Accounts sections are **completely hidden** (not view-only) — only the Preferences section is shown.

10. About

Purpose: System overview, core feature summaries, user-friendly manual reference, and troubleshooting guidance for non-technical users.



About Page Sections

The About page is a **scrollable single-page layout** (not tabbed) with these sections:

Section	Content
Hero	App name, logos (Alumni DB, CEN, SLSU Seal), version number

Section	Content
What is Alumni DB?	Purpose, PTC-ACBET accreditation goals, 4 OBE KPIs
Who is this for?	4 user roles: College Dean, CE/CpE/EE Chairpersons
How It Works	4-step overview: Data Collection → Sync → Review → Report
What You Can Do	8 feature summary cards (Dashboard, Directory, Profiling, Sync, Email, Reports, Settings, Help)
FAQs	Common questions with answers (component: <code>faq-accordion.tsx</code>)
Troubleshooting	Reference to Help page's Troubleshooting tab (8 categories, 31 entries)

9 Core Features Listed

1. Login & Accounts
2. Analytics Dashboard
3. Alumni Directory
4. Alumni Profiling
5. Reports & Export
6. Data Sync
7. Sending Emails
8. Settings
9. About

Troubleshooting Reference

The About page references the **Help page** (</help> → Troubleshooting tab) which contains a comprehensive troubleshooting guide with **8 categories** and **31 problem-cause-solution entries**:

1. Application Startup Issues (3 entries)
2. Login Problems (4 entries)
3. Data Sync Problems (4 entries)
4. Data & Records Issues (4 entries)
5. Email Problems (4 entries)
6. Export & Report Problems (3 entries)
7. Display & Interface Issues (4 entries)
8. Safety & Data Protection (4 entries)

User Manual Design

- **Simple language** — no technical jargon
- **Large numbered steps** — clear action → result format
- **Visual aids** — screenshots/illustrations where applicable
- **FAQs** — common questions with reassuring answers
- **Troubleshooting** — 8 categories of common problems with step-by-step solutions
- **Cross-reference:** Full detailed troubleshooting is available in the Help page (</help> → Troubleshooting tab)

Architecture Summary

Module	Key Characteristics
System Overview	Login gate → role detection → hub-and-spoke with 8 modules
Login & Access	Sheets-based auth, encrypted offline fallback, 3-attempt lockout, 4 roles
Analytic Dashboard	Role-filtered stats, summary cards, % KPIs, freq/WM survey tables, PDF/DOCX export
Alumni Directory	5 actions (Search, Add via GForm, Edit, Delete, View Profile), auto-sync queue
Alumni Profiling	3 display views (History, Latest, Timeline), snapshot-based history
Reports & Export	3 export formats (PDF, DOCX, XLSX) + Print Current View , role filtering
Data Sync	4 sync modes (Pull, Push, Full, Auto-Sync), conflict resolution, unsaved change alerts, all roles can sync
Sending Emails	Compose + History/Inbox branches, role-filtered recipients, GForm link option, 4 ready-made templates
Settings	SMTP + Sheets config, Dean-only management, preferences for all, dark mode, Chairpersons can sync own program via /sync
About	Scrollable page (System Info, Core Features, How It Works, FAQs, Troubleshooting reference), non-technical manual

Changelog (v3.0.0)

Change	Section	Description
Dashboard Export	§3 + §3a	Added PDF/DOCX export branch after charts/tables display. New sub-flow 3a details the export dropdown flow.
Reports Simplified	§6	Removed Print Preview/Preview export type. Now 3 export formats + simple Print Current View.
Email Templates	§8 + §8a	Added ready-made template selection step (4 templates) before compose. New sub-flow 8a details template selection with actual subject lines from code.
Chairperson Sync	§7, §9	Clarified that all roles can access Data Sync module. Settings flowchart updated to show Chairperson sync access via /sync route. Role table corrected: SMTP/Sheets/Sync sections are hidden (not view-only) for Chairpersons.

Change	Section	Description
About Troubleshooting	\$10	Added FAQs and Troubleshooting sections. Cross-references Help page's 8-category, 31-entry troubleshooting guide. Documented scrollable single-page layout.
Architecture Summary	Summary	Updated all modified module descriptions with new capabilities.